

Module 05: Action in Robotics

Learning Objectives

1. Define **Action** within the context of Robotics.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Action.

Introduction

This module explores **Action**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In Robotics, we see Action manifest in: * **Specific Example 1:** A Linear Quadratic Regulator (LQR) controlling a balancing inverted pendulum robot generates motor torques that minimize

a quadratic cost function (weighting position error and control effort), which can be interpreted as an Active Inference action policy where the Q matrix encodes the precision of preferred states (upright position) and the R matrix encodes the cost of acting -- the resulting Riccati-equation solution yields the optimal feedback gain that minimizes expected free energy under linear-Gaussian assumptions. * **Specific Example 2:** An impedance controller on a robotic manipulator performing surface polishing generates torque commands that make the end-effector behave as a virtual mass-spring-damper system, where the "spring" pulls the tool toward the desired contact force (the prior preference) and the "damper" suppresses oscillations; in Active Inference terms, the impedance parameters set the precision of the agent's action model, and the controller continuously acts to minimize the discrepancy between desired and measured interaction forces at 1 kHz.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.